

Chapter 7

Glycan Compositions with GlyConnect Compozitor to Enhance Glycopeptide Identification

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Abstract

Glycoproteomics is unquestionably on the rise and its current development benefits from past experience in proteomics, in particular when attending to bioinformatics needs. An extensive range of software solutions is available, but the reproducibility of mass spectrometry data processing remains challenging. One of the key issues in running automated glycopeptide identification software is the selection of a reference glycan composition file. The default choices are often too broad, and a fastidious literature search to properly target this selection can be avoided. This chapter suggests the use of GlyConnect Compozitor to collect relevant information on glycosylation in a given tissue or cell line and shape an appropriate glycan composition set that can be input in the majority of search engines accommodating user-defined compositions.

Key words Glycan, Glycopeptide identification, Software, Glycosylation, Database

1 Introduction

Glycosylation plays a major role in creating the diversity of proteoforms reaching the magnitude of 10^6 in the human proteome. The mapping of this diversity is far from being charted, particularly in the case of glycoforms, mainly because of the limitations and the very broad variety of experimental techniques needed to capture information. Mass spectrometry (MS), nuclear magnetic resonance (NMR) as part of a fuller range of analytical methods [1] are used to solve the structure of glycans. In recent years, the improved accuracy of mass spectrometers has contributed to refining the identification of intact glycopeptides (peptide with attached glycan), and new glycoproteomics datasets are regularly published.

Glycomics and glycoproteomics have benefitted from past experience in proteomics, and the principles of the Proteome Standards Initiative [2] and ProteomeXchange [3] are being gradually applied. In the past decade, reporting guidelines have been issued

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through the MIRAGE (Minimum Information Required for A Glycomics Experiment) initiative (https://www.beilstein-institut.de/en/projects/mirage/) as part of the many minimum information standards defined to facilitate data checking and processing [4]. Glycomics mass spectrometry data were taken one step further with the introduction of a MIRAGE-compliant experimental data repository named GlycoPOST [5] and the implementation of a pipeline for collecting such data [6]. Some publishers have started imposing data deposition and the field of glycoscience is slowly (re)connecting to Life sciences.

Determining a glycosylation profile may entail either glycomics or glycoproteomics studies or both, depending on experimental design. In most cases, mass spectrometry (MS) data analysis software is used. Despite the expansion of glycan [7] and glycoconjugate databases [8, 9], database search is not an option for glycan identification because there is no direct template that can be used to infer structures detectable in an organism. In glycoproteomics, various strategies are implemented, often centered on peptide identification. These may entail tuning an established proteomics search tool to identify multiple modifications or exploiting idiosyncratic fragmentation patterns of glycans. Subheading 2 of this chapter briefly reviews these aspects.

The explosion of glycopeptide identification software in recent years and the subsequently observed lack of reproducibility led the leaders of the HUPO GlycoProteomics Initiative[1] (HGI) to organize a software challenge. A call was launched in 2017, data was made available in 2018, and the competitors submitted their analyses in 2019. Complex mixtures of N- and O-glycosylated proteins from human serum were processed to produce CID, ETD, and EThcD spectra. Some synthetic glycans were also introduced. At the time of writing, the manuscript describing the results of the competition is still in preparation, but the main authors (Rebecca Kawahara as first and Morten Thaysen-Andersen as last) were kind enough to share their conclusions to feed the discussion of this chapter.

Nine teams of *developers* and 13 teams of *users* entered the competition until the end. Given the diversity of informatics tools and approaches used to identify intact glycopeptides, the HGI challenge organizers mainly remarked on a significant variance of glycopeptide/protein identifications within and between *users* and *developers*. More strikingly, there was a notable discordance between participants using the same software. The spread of identifications was mainly caused by the diversity of search engines, their settings, output filtering and peptide "counting." The experimental design may have dis/advantaged some software and approaches.

[1] https://www.hupo.org/Glycoproteomics-(B/D-GPP)

In the end, 163 common N-glycopeptides were reported by most teams and therefore considered as a high confidence list. Through the comparison of the different submissions, it was concluded that some software solutions perform better than others, and some search strategies yield better results. These observations will undoubtedly guide the improvement of automated glycopeptide data analysis methods.

This chapter briefly surveys the landscape of glycopeptide identification software and focuses on the criteria for selecting a glycan composition file, a key setting significantly impacting the quality of identification. This question is illustrated with the usage of a tool called Compozitor, specifically designed to shape appropriate glycan composition files based on metadata.

2 Glycopeptide Identification Software

2.1 An Ever-Growing Catalog

The topic of automatic intact glycopeptide identification is regularly reviewed [10–13], following the abundant production of new tools that warrant frequent updates of the catalog. This chapter is not destined to add to this set of reviews and, as such, does not span the detailed usage of this type of software. Nonetheless, Table 1 provides the latest coverage of the current range in light of the HGI challenge mentioned in the introduction. Moreover, three methods [14–16] were released recently putting forward the value of using open modification search based on spectral libraries as already achieved in proteomics [17, 18] as well as glycomics [19]. Table 1 also highlights the possibly short lifespan of software tools by ranking the list by date of last software update and distinguishes the glycan types that are possibly identified. Information regarding the impact of using various fragmentation methods is not included, but this feature is properly detailed in [11, 12, 20]. Interestingly, GlycoMod [21], the oldest interface for the manual annotation of intact glycopeptides remains functional 20 years after its release and still efficiently supports small scale studies.

2.2 Glycan Composition File Selection

One of the key points in running glycopeptide identification software is the selection of the glycan composition file upon which the identification of the glycan moiety is based. Table 2 shows the different options offered by tools mentioned in Table 1 and which date of update is less than 4 years old, except GlycopeptideGraphMS [22] for which no composition input is necessary.

Most classical proteomics search engines accommodate the selection of glyco-based modifications from a collection they provide. For example, in principle, Mascot [23] relies on the UniMod subset of glycans [24]. The user may also use in-house definitions as in [25] where a few hundred compositions were built from a

Table 1
Available tools for intact glycopeptide MS data analysis sorted by date of last update

Software name	Web interface	Download site		Latest update	References
O-pair		https://github.com/smith-chem-wisc/MetaMorpheus	O-linked	11/2020	[14]
SugarPy		https://github.com/SugarPy/SugarPy	N/O-linked	10/2020	[16]
Protein prospector	https://prospector.ucsf.edu/prospector/mshome.htm		N/O-linked	09/2020	[35]
MS Fragger Glyco		https://msfragger.nesvilab.org/tutorial_glyco-fragger.html	N/O-linked	09/2020	[15]
GlycoReSoft		https://github.com/mobiusklein/glycresoft	N/O-linked	04/2020	[36]
pGlyco		http://pfind.ict.ac.cn/software/pGlyco/index.html#Downloads	N-linked	01/2020	[37]
Glycopeptide GraphMS		https://bitbucket.org/glycoaddict/glycopeptidegraphms/src/master/	N/O-linked	12/2019	[22]
glyXtoolMS		https://github.com/glyXera/glyXtoolMS	N/O-linked	10/2018	[38]
GlycoProteome Analyzer (IQ-GPA)		https://www.igpa.krDiscontinued?	N/O-linked	10/2017	[39]
SugarQb		http://ms.imp.ac.at/?action=sugarqb	N-linked	09/2017	[40]
GlycoPAT		https://virtualglycome.org/glycopat/	N-linked	01/2017	[27]
MAGIC	http://magic.iis.sinica.edu.tw/MAGIC.php	http://ms.iis.sinica.edu.tw/comics/software_magic	N/O-linked	08/2016	[26]
GlycoSeq		https://github.com/chpaul/GlycoSeq/	N-linked	06/2016	[41]
GlycopeptideID	http://glycopeptideid.appliednumerics.fi/		N-linked	11/2014	[42]
GPFinder		https://github.com/gpfinder/gpfinder	N/O-linked	10/2014	[43]

(continued)

Table 1
(continued)

Software name	Web interface	Download site		Latest update	References
GlycoPep ID/Evaluator	http://hexose.chem.ku.edu/predictiontable.php	https://desairegroup.ku.edu/research	N-linked	2014	[44, 45]
GlycoPeptideSearch		https://edwardslab.bmcb.georgetown.edu/trac/GlycoPeptideSearch	N-linked	04/2013	[46]
GlycoFragWork		http://darwin.informatics.indiana.edu/col/GlycoFragwork/download.html	N-linked	03/2013	[47]
Sweet-Heart		Only available on request	N-linked	2009	[48]
Sweet-NET		Not available	GAGs	NA	[49]
GPQuest		Only available on request	N-linked	NA	[50]
GlySeeker		Only available on request	N-linked	NA	[51]
GlycoMaster DB		Discontinued?	N-linked	?	[52]
Byonic		Licensed	N/O-linked	NA	[53]

Table 2
Parametrization of intact glycopeptide search engines with glycan compositions

	Tool name	Theoretical definition	Empirical definition	Customizable definition
Proteomics search engines adapted to glycosylation	MASCOT	✓		✓
	ProteinProspector		✓	
	SeQuest	✓		✓
	MS Amanda	✓		✓
Dedicated glyco-proteomics search engines	SugarQb	✓		
	GlyxTools	✓		✓
	GlycoPAT	✓		✓
	O-pair		✓	
	MS Fragger Glyco		✓	✓
	pGlyco		✓	
	GlycoReSoft	✓		✓
	SugarPy	✓		
	MAGIC	✓		✓
	GlycoSeq			
	IQ-GPA			
	Byonic		✓	✓

customized and systematic addition of monosaccharides prior to using the Mascot engine.

In many tools, the composition file is composed of glycan data collected from the literature or from databases/repositories. This is qualified as empirical in Table 2. Some software use the knowledge of glycan biosynthesis to generate potential glycans. For example, MAGIC [26] starts with 29 monosaccharides to be combined according to enzymatic rules and in a similar way, 19 groups of seed structures are defined in GlycoPAT [27]. In this way, the composition file is "theoretically" defined as the result of potential enzymatic activity but flexible as both the enzymes or the monosaccharides can be removed or constrained. SugarPy [16] takes all possible combinations of monosaccharides.

Note that instrument vendors also provide a glycopeptide identification component in the software suites that are usually available for proteomics data analysis such as ProteomeDiscover by Thermo or ProteinScape at Bruker, but the information on glycan composition files is not easily accessible.

Proteomics software developers, in particular when posttranslational modifications are accounted for, have already acknowledged that the performance of search engines correlates with limiting the search space [28]. Similar observations have been made in glycoproteomics, and this viewpoint, also promoted here, is strongly advocated in [29].

The remainder of this chapter is dedicated to tailor-making a glycan composition file. In preamble, it should be stressed that there is no standard notation to express glycan composition as reported in [30]. We rely here on a so-called condensed notation: H for hexose, N for hexosamine, F for fucose, and S for sialic acids. Additionally, small letters designate modifications such as "a" for acetylation, "p" for phosphorylation, and "s" for sulfation, for the most frequent.

3 Practical Examples

The following describes detailed procedures to explore glycome data and derive an appropriate set of glycan compositions to be considered as usable input in the software listed as accommodating custom definitions in Table 2. It involves using GlyConnect Compozitor [30], a tool specifically designed to map compositions related by shared monosaccharide counts into a graph that eases glycome visualization and comparison. When available, each composition is associated with defined structures. The Compozitor tool is integrated in the GlyConnect database where data and information on glycosylation, glycoconjugates, and glycans are manually curated. All cited resources that are not in Table 1 are listed in Table 3 with summary information.

Table 3
Summary of cited glycoinformatics resources

Resource name	Type	URL	Reference
GlyConnect	Database	https://glyconnect.expasy.org	[8]
Compozitor	Software	https://glyconnect.expasy.org/compozitor	[30]
GlycoMod	Software	https://web.expasy.org/glycomod	[21]
GlycoPOST	Repository	https://glycopost.glycosmos.org	[5]
UniMod	Database	http://www.unimod.org	[24]

3.1 Exploring the Urine O-Glycome in GlyConnect

From the GlyConnect homepage:

1. Select the TISSUE button to open the Tissue View of the database.
2. In the Tissue View page, type "urine" in the search window.
3. In the only output row, click on the "urine" to open the corresponding page (tissue ID: 31).
4. The "urine" entry page shows a broad variety of glycan structures that can be better summarized by clicking on the Compozitor link on the right side.
5. The Compozitor search fields are all pre-filled in the "Source" tab with details of the tissue entry. In this case, the tissue name is "Urine" and the species name is set by default to "*Homo sapiens*" (changing this parameter is straightforward in the Species field).
6. In order to focus on the O-glycome, deselect "N-linked," "C-linked," and "Free" options in the glycan-type field. When this is done, click on the "Add to selection" button. This will confirm that 75 O-glycan compositions have been extracted from the database.
7. Clicking on the "Compute graph" button will display the graph of all glycans reported as O-linked on urinary proteins and a bar plot that reflects the main characteristics of the glycans.

   This output is shown as the left-hand side of the graph in Fig. 1, labeled A. Each node is a glycan composition labeled in a condensed notation (*see* Subheading 2). Two nodes are connected if they share a composition with only one monosaccharide difference. The differing monosaccharide encoding letter labels the edge. The number featuring in the node summarize the number of defined glycan structures matching the composition and stored in the database in relation to the queried tissue.

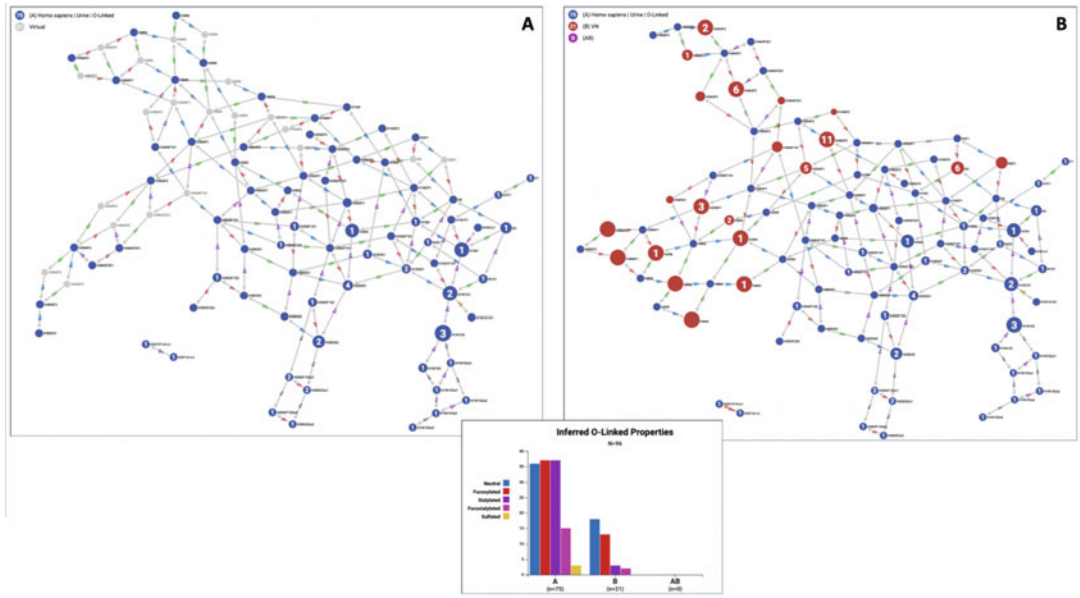


Fig. 1 Compozitor output of the human urine O-glycome. (**a**) Graph of all O-glycome glycan compositions, reflecting the content of the GlyConnect database on human urine specifically, and where virtual nodes (see text) are displayed in gray. (**b**) Same graph highlighting virtual nodes in red and reflecting data retrieved from the entire GlyConnect database. The node label corresponds to the number of structures matching the composition in the database and the node size, to the number of references citing the composition. The bar chart displays the distribution of five general properties of O-glycans across the urine O-glycome and across the set of virtual nodes (*see* **Notes 1, 3** and **4**)

Compozitor highlights the consistency of a glycome by showing relatedness between glycan compositions, hence, structures. In the case of human urine, the central part of the graph is tightly connected. Importantly, Compozitor allows for "virtual nodes" (in gray) when only one intermediary step is needed to connect isolated nodes. If not for these, the graph would be more scattered especially the left side of the graph 1A where the virtual nodes are dominant. In fact, compositions with more than three hexoses are scattered. Virtual nodes can be considered as potentially missed structures in experimental results as discussed in [30]. This point will be further developed here and in the next section.

The output graph shows the number of defined structures associated with each composition, thereby reflecting the database content. It appears that selected compositions have one to three possible matches in the database. Compozitor is also suggesting to resort to adding 21 virtual nodes to fully connect the graph. In fact, these nodes are only virtual as a reflection of the content of experimental datasets. Refining information in

the graph is recommended in order to test the realism of these additional nodes.

8. In the page updated at **step 7**, click on the "Export" button below the graph and select "Virtual" only (i.e., deselect all other options). Then, click on the clipboard icon to copy the selection of 21 compositions.
9. Select the "Custom" tab of Compozitor and paste the selection of 21 compositions in the query window. Select O-linked as a glycan type in the corresponding window. Optionally, name the list as, for example, VN. Click on the "Add to selection" button and the "Compute graph" button.

The resulting graph is shown in Fig. 1b. All virtual nodes are now displayed as red nodes. By using the "Custom" tab of Compozitor and accessing the content of the entire GlyConnect database reveals that compositions corresponding to the virtual nodes are actually mapped with the existing defined structures as highlighted in the node labels of Fig. 1b. The size of the nodes currently reflects the number of references stored in the database and reporting this composition. The number label as in all other cases is the number of corresponding structures stored in GlyConnect. Figure 1b shows that most of the virtual nodes have been mentioned in stored references, and this would suggest that 75 + 21 compositions should be used as a representative set for identifying glycopeptides in human urine samples. The associated bar chart shown in Fig. 1 is generated by Compozitor to highlight the overall glycan properties of the urine O-glycome and comparatively of the virtual node contribution. Additional fucosylated glycans are suggested. Composition data as well as their graphic representation (s) can be easily downloaded by clicking on the green "Export" button mentioned at **step 8** (*see* Subheading 5).

3.2 The N-Glycome of Erythropoietin

The glycosylation of erythropoietin (EPO) has been extensively studied over the years, and the GlyConnect database stores 18 references, two of which are glycoproteomics data [31, 32].

To view and study the N-glycome of EPO, from the GlyConnect homepage, proceed as follows:

1. Select the PROTEIN button to open the Protein View of the database.
2. In the Protein View page, type "poietin" in the search window.
3. In the output of 10 entries, select entry 140 (Id column), i.e., erythropoietin, for display.
4. On the right side of the erythropoietin entry page, click on the Compozitor link.
5. The Compozitor search fields are pre-filled with the details of the 140 entries. In this case, the protein name is

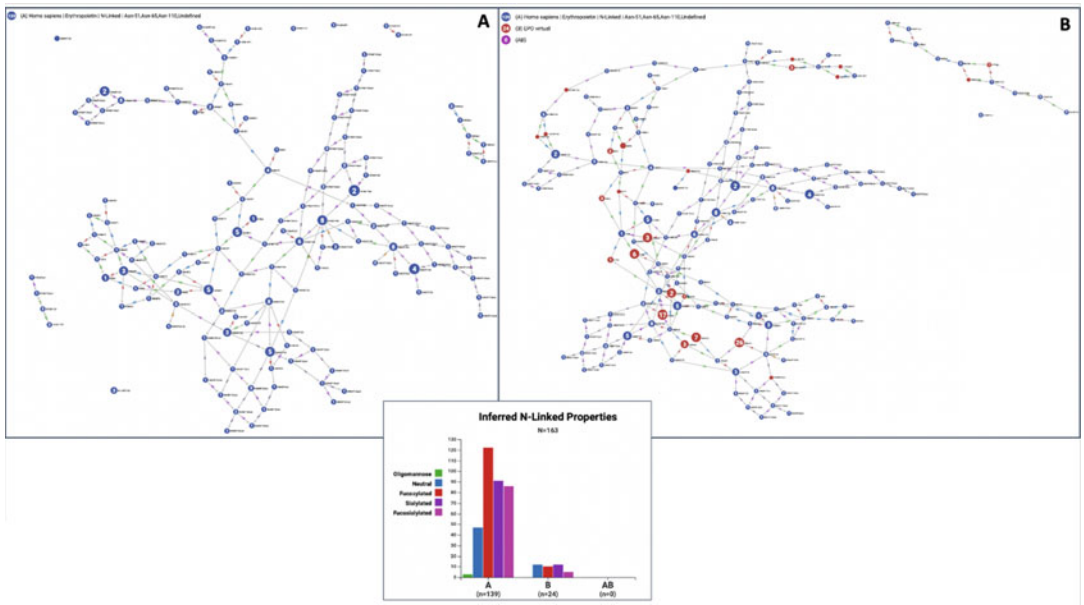


Fig. 2 Compozitor output of the N-glycome of human EPO. (**a**) Graph of all N-glycome glycan compositions reflecting the content of the GlyConnect database on human erythropoietin specifically. No virtual node is included in this case. (**b**) Same graph highlighting virtual nodes in red and reflecting data retrieved from the entire GlyConnect database. The node label corresponds to the number of structures matching the composition in the database and the node size, to the number of references citing the composition. The bar chart displays the distribution of five general properties of N-glycans across the N-glycome of human EPO and across the set of virtual nodes (*see* **Notes 1, 3** and **4**)

"erythropoietin" and the Species is "*Homo sapiens*," shown in the "Protein" tab.

6. In order to focus on the N-glycome, deselect "O-linked," "C-linked," and "Free" options in the glycan-type field. When this is done, click on the "Add to selection" button and display the graph of connected compositions by clicking on the "Compute graph" button.
7. The displayed graph shows a well-connected set of 139 compositions representing the protein N-glycome as well as a bar plot showing the main characteristics of the glycans. In this case, there is an observable bias toward fucoslyated structures.

   Figure 2a on the left shows the graph generated after deselecting the inclusion of virtual nodes to expose the scattering of nodes. Figure 2b shows the graph after inclusion of the virtual node subset (as per method described for urine in Subheading 3.1 with the need for selecting "N-linked" as the only difference). Virtual nodes are shown in red. Figure 2 also shows the bar plot of glycan properties highlighting the dominance of fucosylated structures.

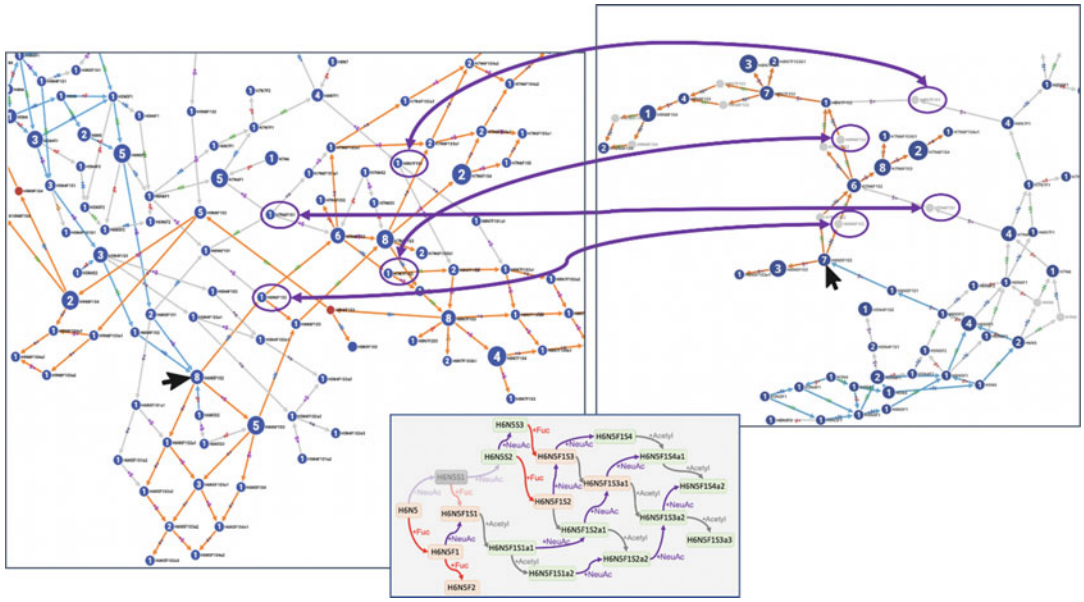


Fig. 3 Mapping old and new data with GlyConnect Compozitor. Two Compozitor graphs of EPO N-glycome data are compared. On the right panel, the graph is the result of processing a dataset of 58 compositions from 17 references, and the left panel shows the result with 139 compositions from 18 references. The purple arrows link the nodes that were virtual (see text) in the right graph and became identified compositions in the left graph. The middle panel shows the connectivity of the H6N5 composition: pink nodes are compositions found in both the right and left panels, and green nodes are exclusively in the left panel. Each node is connected to another through either fucoslyation, sialylation, and acetylation. Almost all possible options are mapped in this graph except for H6N5S1. This composition is either below the detection threshold in the corresponding experiments or absent for a reason to be investigated

Because EPO is particularly popular, it was selected as an illustrative example in the presentation of Compozitor in [30] but at the time, the 18th reference associated with it [32] was not yet integrated. As it turns out, the inclusion of new data validates our virtual node approach. Figure 3 shows the graph comparison between the mapping of 17 references as depicted in [30], on the right and the mapping of 18 references realized for this chapter and described above, on the left. The original figure was centered on H6N5F1S2 showing outgoing paths in orange and incoming paths in sky blue. Only an excerpt is shown in Fig. 3. The same was applied to the current graph to ease comparison despite two very different topologies.

One of the challenges of defining virtual nodes is the impossible choice between two concurrent options. In our example, to connect H6N5F1S2 with H7N6F1S2 there is no reason to favor either H6N6F1S2 or H7N5F1S2 without further information. That is why in many cases, two paths through two virtual nodes are proposed by Compozitor. The inclusion of new data often provides the missing links as illustrated in

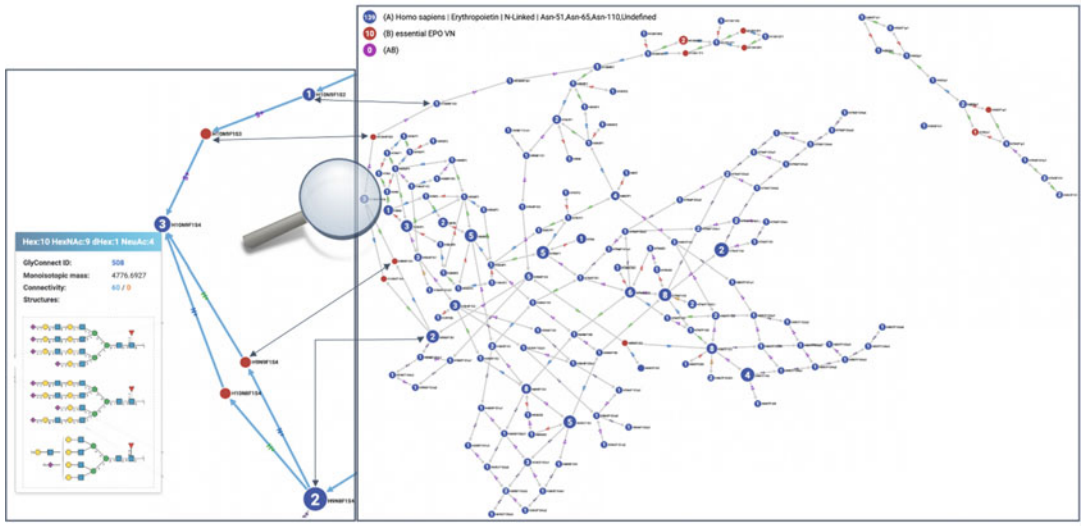


Fig. 4 The EPO N-glycome with reduced virtual nodes. The graph of Fig. 2b was reduced by 14 virtual nodes, the removal of which did not disrupt the graph topology. It is shown in the right panel. A close-up of H10N9F1S4 is shown on the left highlighting the isolation of this node if not for the red virtual nodes on both sides. This begs the question of why are the H10N9F1S3 and H10N8F1S4/H9N9F1S4 compositions not present in the experimental results (*see* **Notes 1, 3** and **4**)

Fig. 3. The most recent graph on the right has turned some virtual to real nodes and also suggests the obsolescence of earlier options. There is now a reason to favor H6N6F1S2 over H7N5F1S2 simply because the former was mapped while the latter was not. Likewise, H8N6F1S2 is supported over H7N7F1S2 to connect H7N6F1S2 with H8N7F1S2. Figure 3 also highlights the relevance of another two virtual nodes: H7N6F1S1 and H8N7F1S1 that provide intermediary monosialylated forms of the proven disialylated H7N6F1S2 and H8N7F1S2. Lastly, a supplementary graph showing all compositions stemming from H6N5 as the result of fucoslyation, sialylation and acetylation in EPO is shown in the last panel of Fig. 3. It brings out the absence of monosialylation independent of fucosylation. Whether H6N5S1 should be included as possible composition is an open question.

The 24 virtual nodes were reconsidered, and 14 of them could be removed without breaking the network of EPO N-glycome. In other words, the remaining 10 nodes are key to holding the topology of the graph together. The graph with these 10 structurally essential virtual nodes is shown in Fig. 4 with a close-up on H10N9F1S4 that is included in the graph only because of the inclusion of three virtual nodes. It would seem logical to account for these intermediary steps in an attempt to identify EPO glycopeptides as performed in [32]. Note that the graph was slightly modified from the raw

output to help the readability of the figure. The interface makes it easy for the user to move the nodes around (*see* **Note 2**).

Further exploration may entail comparing the individual glycomes at distinct glycosylation sites. In the case of EPO, the procedure continues as follows:

8. Click on the cross next to "Selection A" to remove the selection of all N-linked sites.
9. Deselect all sites by clicking on the minus sign right before the "Site" field.
10. Select site Asn-51 and then click on the "Add to selection button."
11. Deselect Asn-51, select site Asn-65 and click on the "Add to selection button."
12. Deselect Asn-65, select site Asn-110 and click on the "Add to selection button."
13. Click on the "Compute graph" button to display the comparative graph where composition nodes will be distinguished by color: blue for Asn-51, red for Asn-65 and green for Asn-110.

Figure 5 shows the result. As can be expected, the compositions that are common to all sites tend to be central in the graph while the

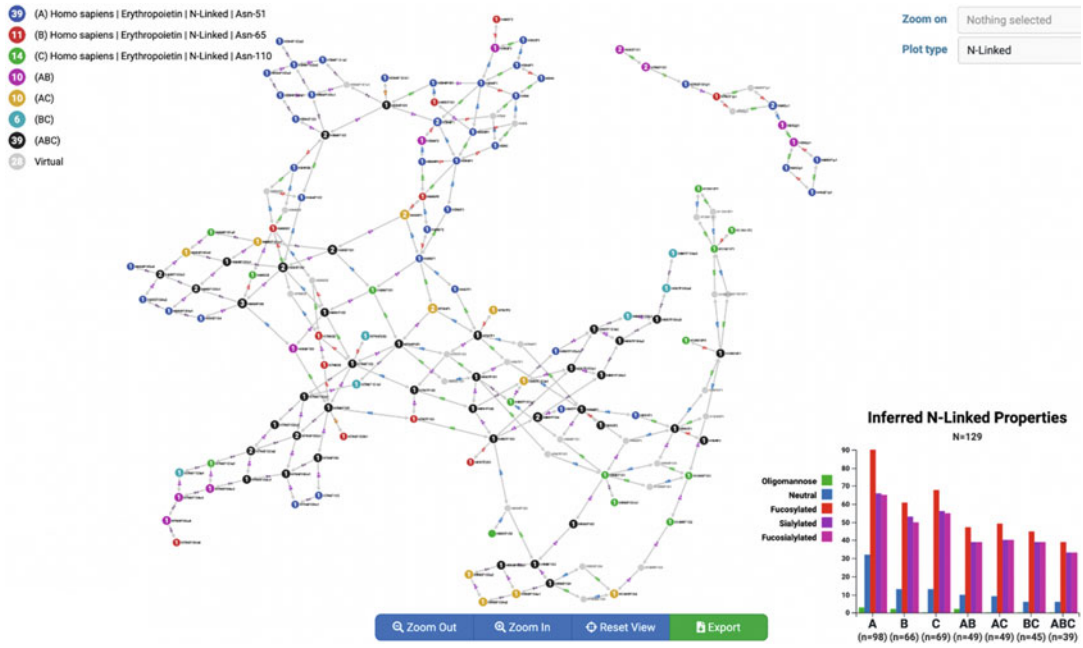


Fig. 5 Compozitor output for glycosite-specific N-glycomes in human EPO. Upon selection in the query window, GlyConnect Compozitor can output the complete set of glycan compositions associated with a single glycosite in a glycoprotein. This operation can be repeated with an alternate glycosite and the two resulting graphs are combined. The graph reveals that the N-glycomes of Asn-51, Asn-65, and Asn-110 are distinguishable. The bar chart summarizes the glycan properties. For example, there is no oligomannose and proportionally more sialylated compositions on Asn-110 in comparison with the other two sites (*see* **Notes 1, 3** and **4**)

specific structures produced through the addition of various monosaccharides are found as leaves. Note that the comparative display is currently limited to up to three entity selection, whether proteins, sites or others (*see* [30] for further examples and **Note 5** for further options).

3.3 Using GlyConnect Metadata

The GlyConnect database can be used to refine information collected in a Compozitor graph. Each node in the graph is cross-linked to a composition page summarizing the database content with respect to that particular composition. When available, information on possible structures is displayed. For example, in the urine O-glycome shown in Fig. 1, the H3N3F1S2 node is labeled 1, meaning that one urine-related glycan structure is known. Note that most surrounding nodes are not labeled. Mousing over the node prompts an information caption containing the composition ID link (444 for H3N3F1S2), which if activated, opens a new page dedicated to information relative to H3N3F1S2. Access to this page and the page itself are shown in Fig. 6. GlyConnect data access is supported through a series of clickable buttons, the color of which follows an established code described in [8]. Briefly, green

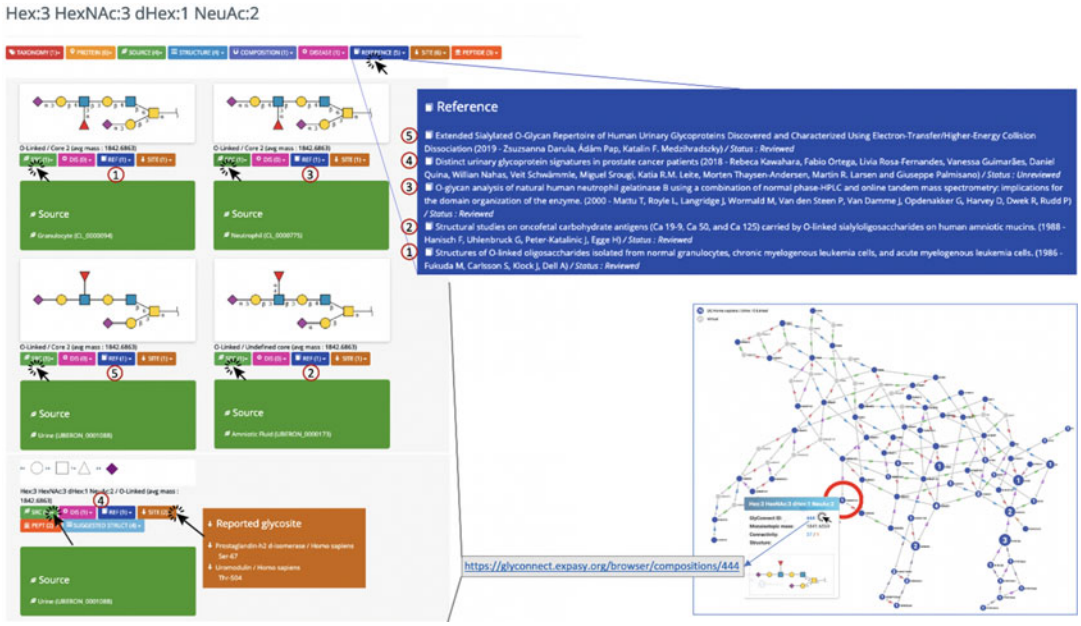


Fig. 6 Use of glycan structure metadata. A node is singled out in the Compozitor graph of the human urine O-glycome of Fig. 1a reproduced on the bottom right, for further investigation. Mousing over the node displays associated structures and a link to the corresponding composition summary page in GlyConnect can be activated to open that page. The latter is displayed on the left. Metadata can be checked by clicking on colored buttons. The green (resp. brown) buttons reveal the tissue sources (resp. protein name and sites). The information shown confirms that the position of the fucose residue (red triangle) differs between blood cell lines and body fluids

is for tissue or cell line information, brown for site-specific information, and electric blue for references. Clicking on the reference button opens the list of publications in reverse chronological order. According to the GlyConnect content, H3N3F1S2 is mentioned in five articles published between 1986 and 2020, numbered 1–5 in Fig. 6. Four of these references report more or less precise glycan structures, and the fifth article that describes a large-scale glycoproteomics experiment specifies glycopeptides carrying the H3N3F1S2 composition. Interestingly and as a representative example of a usual situation, precisely defined structures tend to be found in older publications with low-throughput technology, whereas structures with missing information (mainly on linkages) are released in more recent articles relying on high-throughput MS-based methods.

The striking point of Fig. 6 is the position of the fucose residue. In refs. [1] and [3] where the glycan is expressed in two hierarchically related cell types (granulocyte and neutrophil), fucose is $(\alpha - 3)$ linked to a GlcNAc residue. In refs. [2] and [5] reporting glycan expression in body fluids (urine and amniotic fluid), fucose is $(\alpha - 4)$ linked to a GlcNAc residue. This clear and meaningful contrast may guide users in interpreting their own data depending on tissue expression. In particular, when checking glycan compositions resulting from a glycoproteomics experiment as the one described in ref. [4], H3N3F1S2 seems more likely to match the structure described in refs. [2] and [5], based on tissue information. This possible match remains a suggestion that can be substantiated with further experiments.

Figure 6 also shows that clicking on the site brown button associated with a structure reveals the identity of the protein carrying the glycan, when known and the position(s) of its glycosite(s). The H3N3F1S2 composition found in the glycoproteomics experiment of reference [4] was mapped on two distinct sites of two distinct proteins, namely uromodulin (P07911) and prostaglandin-H2 D-isomerase (P41222). Unfortunately, the more precise structure reported in ref. [5] was found on Protein YIPF3 (Q9GZM5). In that case, independent sources of information are not corroborating.

4 Conclusion

Glycoproteomics is rapidly expanding and likely to produce large volumes of data in the near future. In particular, comparative studies such as [33, 34] reveal the subtlety of site-specific glycosylation and profile variation in glycan expression. Capturing this information and using software tools to display and compare the profiles at the site level is the next frontier in sight. The GlyConnect database will evolve accordingly, and Compozitor is geared to

tackle quantitative data that could be expressed as the size of a node, currently reflecting the number of citing references. Compozitor was designed to move away from presenting lists of identified glycan compositions as if each item was independent of each other. This tool provides an overview of glycome consistency, and the more data in the database, the more reliable this representation will become. At this point in time, there is unfortunately an imbalance in data production being biased toward human. Compozitor is therefore more likely to provide a comprehensive glycan composition file for customization of the corresponding parameter in glycoproteomics search engines. Nonetheless, this may not completely exempt the user from searching literature in the case of studies with species less popular than mammals. Being aware of this limitation is definitely an incentive to attempt to reduce the species gap in the database.

5 Notes

1. Several of the images showing a Compozitor graph in figures have been slightly modified from the raw output to disentangle the network which is generated by the D3.js (https://d3js.org) library. This transformation is made easy since any node can be dragged wherever a user wishes in the space of the browser window by maintaining the mouse over it. Consequently, paths can be shortened or stretched upon user actions.
2. Compositions can be exported in a text file as soon as they have been added to the selection and before computing the graph ("Export selection" button). They can be selectively exported once the graph is computed ("Export" button). The graph can be exported in the .svg format.
3. The "Zoom in" and "Zoom out" buttons have obvious purposes, and the former operation is supplemented by the "Zoom on" window located in the top right corner; typing a specific composition in this window will trigger zooming in and centering the graph on the corresponding node.
4. Mousing over a bar of the bar plot of glycan properties highlights all nodes that are counted in that bar to show where they are located in the graph.
5. Compozitor offers six tabs for querying GlyConnect and build composition files. The main text illustrated the use of the "Protein", the "Source" and the "Custom" tabs. Other options include searching by cell line as well as by disease. In the last tab labeled "Advanced," queries directly launch programmatic access to GlyConnect via its API (Application

Programming Interface). For example, typing the following query in the only window of this tab:

"taxonomy=homosapiens&glycanType=N-linked&tissue=blood serum&disease=carcinoma" has the same effect as filling the corresponding fields (selecting *Homo sapiens* in "Species," blood serum in "Tissue," and N-linked in "Glycan type") in the "Source" tab and in the "Disease" tab (selecting *Homo sapiens* in "Species," carcinoma in "Disease," and N-linked in "Glycan type"). In other words, it provides in one step a result that would require two steps.

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